Perceptual Quality Assessment for Light Field Image Compression: A Subjective Database and Objective Metric

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Abstract

Light field images (LFIs) capture rich angular and spatial information, enabling immersive scene visualization. However, their substantial storage requirements necessitate compression, which inevitably introduces perceptual distortions. Despite the emergence of numerous advanced compression methods, existing quality assessment frameworks for LFIs remain narrow in scope, leaving the perceptual impact of diverse compression distortions largely unexplored. To address this, we conduct subjective experiments and present a new perceptual quality assessment database comprising 1000 distorted LFIs from 20 scenes, generated using 10 compression methods across five quality levels, with mean opinion scores collected from 29 qualified observers under controlled conditions. Building on this database, we propose a novel objective blind metric based on multi-modal and multi-task learning. The metric extracts 2D and 3D multi-modal patches from 4D LFIs, feeding them into image-domain and video-domain branches for preliminary quality assessment. A distortion-aware multi-task learning module then integrates cross-modal features to produce the final quality score. Extensive experiments on the proposed database demonstrate that the proposed metric outperforms representative quality assessment metrics in various aspects. The database and metric code are available at https://github.com/oldblackfish/HUST-LFI.

CCS Concepts

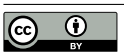
• **Human-centered computing** → **Visualization design and evaluation methods**.

Keywords

Light field image quality assessment, Database, Blind metric

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1 Introduction

In the rapidly evolving era of visual communication, conventional 2D images are increasingly unable to meet the growing demand for immersive visual experiences. Light field imaging, as an innovative and emerging technology, is capable of simultaneously recording both the angular and spatial information of light rays, thereby providing richer scene representations. Light field images (LFIs) generated by this technology can effectively capture the 3D geometric structure and illumination information of a scene, offering users a more immersive visual experience. As a result, they have attracted extensive attention in applications such as virtual reality (VR) [8], 3D displays [3], and novel view synthesis [20].

Unlike conventional 2D images, the LFI is a high-dimensional imaging representation used to record the distribution of light rays. It was initially modeled as a 7D light field function. For practical acquisition and application, it is commonly simplified into a 4D representation [22] and described using the two-plane parameterization model $L(U, V, H, W)$, where (U, V) denote the angular plane and (H, W) represent the spatial plane. Because LFIs encode both spatial and angular information, they have substantially larger data volumes than 2D images, posing significant challenges for storage and transmission. This has driven the development of various LFI compression methods [9, 41]. However, these methods inevitably introduce distortions distinct from those in conventional 2D images, thereby degrading perceptual quality. Light field image quality assessment (LFIQA) is therefore essential for quantifying compression-induced distortions, providing perceptual feedback, and optimizing compression strategies.

Generally, LFIQA research can be divided into two categories: subjective experiments and objective metrics. Subjective experiments aim to explore the underlying quality relationships among LFIs with different distortions and to establish standard databases for benchmarking objective quality metrics. To date, considerable efforts have been devoted to conducting subjective studies to investigate the impact of compression and reconstruction algorithms on LFI distortions [14, 31, 33, 36, 42, 53]. However, these existing databases suffer from several critical limitations:

(1) **Lack of distortion samples generated by learning-based and light field-specific codecs.** Although compression has become an important research area in light field imaging, most existing datasets still rely primarily on traditional compression methods to generate distorted samples, failing to cover distortions introduced by emerging learning-based methods and light field compression

standards. This limited diversity restricts both the training and evaluation of objective quality assessment models.

(2) **Severely limited dataset scale.** It is well known that high-performing models require sufficient training data. However, most existing databases contain only a few hundred samples, which is insufficient for training robust objective LFIQA models.

(3) **Lack of standardized databases and evaluation benchmarks.** These limitations hinder the development of standardized LFIQA databases and objective metrics. A unified benchmarking framework is therefore needed to systematically assess existing metrics and their effectiveness in practical scenarios.

To address these limitations, we conduct a subjective study using the double stimulus impairment scale (DSIS) protocol and construct HUST-LFI, a database dedicated to compressed LFIs. It comprises 10 real-world and 10 synthetic reference scenes, compressed at five levels using 10 traditional and learning-based image, video, and LFI codecs. Each distorted LFI is assigned a mean opinion score (MOS), yielding 1,000 samples with broad content and distortion diversity.

Based on this database, we further propose a novel objective blind LFIQA metric, which can effectively predict the perceptual quality of LFIs without requiring reference images. Finally, extensive experiments are conducted on the constructed database, and a comprehensive benchmark evaluation is provided to compare the proposed metric with several representative 2D image quality assessment (IQA) and LFIQA metrics. In summary, the main contributions of this paper are as follows:

(1) **Subjective LFIQA Database:** We construct HUST-LFI, which is, to the best of our knowledge, the largest perceptual LFIQA database with subjective quality scores to date. The database contains 20 reference light field scenes, and a set of light field compression distortion schemes based on both traditional and learning-based methods is systematically designed. In total, 1000 LFIs with different compression distortion types and levels are generated.

(2) **Objective LFIQA Metric:** We propose a novel multi-modal and multi-task learning based blind LFIQA metric, termed MM-BLFQA. The proposed metric transforms the 4D light field into multi-modal representations, i.e., 2D MLI and 3D PV, respectively. Furthermore, a dual-branch architecture is designed to predict preliminary quality scores from the respective modalities, where a distortion-aware multi-task learning mechanism provides auxiliary supervision, and preliminary scores from the two branches are adaptively weighted to obtain the final quality score.

(3) **Performance Evaluation:** We establish a comprehensive benchmark on the proposed subjective LFIQA database, systematically comparing the proposed metric with representative 2DIQA and LFIQA metrics. Experimental results reveal the limitations of existing metrics in handling compression-distorted LFIs and demonstrate the superiority of the proposed metric in multiple aspects with relatively low computational complexity.

2 Related Work

2.1 Subjective Databases for LFIQA

Several publicly available LFIQA databases have been established through subjective MOS collection, as summarized in Table 1 and reviewed below in chronological order.

Table 1: Comparative overview of subjective LFIQA databases.

Database	Scale	Distortion Type	Distortion Category	No. SRCs
SMART [31]	256	JPEG, JPEG2000, HEVC, SSDC	Traditional compression	16
VALID [42]	100	HEVC, VP9 Zhao, Ahmad, Tabus	Traditional compression Traditional LFI compression	5
Win5-LID [36]	220	HEVC, JPEG2000 LN, NN, EPICNN, LF-SYN	Traditional compression Reconstruction	10
NBU-LF1.0 [14]	210	NN, BI, VDSR, EPICNN, Zhang	Reconstruction	14
SHU [33]	240	JPEG, JPEG2000 Gaussian blur, White noise, Motion blur	Traditional compression Acquisition or display	8
IETR-LFI [53]	120	HDDRNet, LFASR, LF-IINet, DistgSSR	Reconstruction	10
HUST-LFI (Proposed)	1000	JPEG2000, HEVC, VVC Cheng, ELiC, DCVC-FM, DCVC-RT JPEG Pleno SADN, LFIC-DRASC	Traditional compression Learned compression Traditional LFI compression Learned LFI compression	20

SMART database was first constructed by Paudyal et al. [31] as the earliest database for LFIQA. It contains 16 reference LFIs of real scenes captured using the second-generation light field camera, Lytro Illum. Four compression distortions (JPEG, JPEG2000, HEVC, and SSDC) are applied to each reference LFI at four distortion levels, resulting in 256 distorted LFIs.

VALID database was proposed by Viola et al. [42]. It comprises 5 reference LFIs and 100 distorted LFIs generated using five compression schemes, including two video coding methods (HEVC and VP9) and three light field compression algorithms proposed by Zhao [56], Ahmad [1], and Tabus [39], each applied at four distortion levels.

Win5-LID database was introduced by Shi et al. [36] to investigate subjective light field quality under windowed 5DoF viewing conditions. It includes 10 reference LFIs with four traditional distortions (HEVC, JPEG2000, LN, and NN) at five levels, and two deep learning-based distortions (EPICNN [44] and LF-SYN [19]) with one level each, yielding 220 distorted LFIs in total.

NBU-LF1.0 database was proposed by Huang et al. [14] with a focus on reconstruction distortions. It includes five distortion types, namely NN and BI for angular reconstruction, VDSR [21] for spatial super-resolution, and EPICNN [44] and the method of Zhang [51] for light field angular reconstruction. The database contains 14 reference LFIs and 210 distorted LFIs.

SHU database was constructed by Shan et al. [33], containing 8 reference LFIs and 240 distorted LFIs. It includes 5 distortion types, where JPEG and JPEG2000 correspond to compression distortions, while Gaussian blur, white noise, and motion blur are related to the acquisition or display process, each with 6 distortion levels.

IETR-LFI database was proposed by Zhang et al. [53]. The database contains 10 reference LFIs and 120 distorted LFIs, generated by individually or jointly applying two angular reconstruction methods (HDDRNet [28] and LFASR [18]) and two spatial super-resolution methods (LF-IINet [26] and DistgSSR [43]).

Overall, existing LFIQA databases remain limited. SMART, VALID, Win5-LID, and SHU primarily cover traditional compression, with limited support for learning-based and light field-specific methods; VALID also suffers from limited content and outdated codecs. NBU-LF1.0 and IETR-LFI focus mainly on reconstruction artifacts. In contrast, HUST-LFI covers diverse compression distortions, particularly state-of-the-art learning-based image, video, and LFI compression methods, offering greater scale and diversity.

2.2 Objective Metrics for LFIQA

According to the availability of reference LFIs, objective LFIQA metrics can generally be categorized into three types: full-reference (FR), reduced-reference (RR), and no-reference/blind (NR/B) metrics. Among them, blind LFIQA metrics are capable of directly evaluating quality without requiring reference LFIs, making them more practical where reference light field data are often unavailable. Motivated by this practical demand, this paper primarily focuses on the research progress of blind LFIQA metrics.

Existing blind LFIQA metrics can be broadly divided into two categories: handcrafted feature-based metrics and deep learning-based metrics. Early studies mainly relied on manually designed feature extraction strategies [37, 47, 53, 60], which evaluate the perceptual quality of distorted LFIs by separately modeling the spatial and angular information of light fields. Different from these metrics, Xiang et al. [46] and Zhang et al. [52] extended quality analysis to the frequency domain using four-dimensional (4D) transforms, extracting features directly from the joint spatial–angular structure of distorted LFIs for quality prediction.

In recent years, with the rapid development of deep learning, data-driven blind LFIQA metrics have gradually become the mainstream. These metrics can automatically learn discriminative features and establish nonlinear mappings between distorted LFIs and subjective quality scores. DeeBLiF, proposed by Zhang et al. [54], was the first to adopt a two-stream CNN architecture to disentangle the spatial and angular features. Building on this idea, PVBLiF [55] reconstructs light field data into pseudo-video sequences and incorporates a saliency-guided pooling strategy to enhance the representation of quality-related regions. Subsequently, FSFNet, proposed by Xiang et al. [45], learns spatial–angular quality-aware representations through a self-supervised learning framework, enabling more effective distortion perception. In addition, MAFBLiF, proposed by Zhou et al. [58], introduces a multi-scale attention fusion mechanism to integrate spatial and angular information across different scales, further improving prediction performance.

Despite numerous objective LFIQA metrics, their effectiveness under compression distortions remains unclear due to the lack of a dedicated database, motivating systematic benchmarking on HUST-LFI. Moreover, existing metrics represent 4D LFIs using low-dimensional 2D or 3D single-modal inputs, limiting their ability to capture high-dimensional quality-aware features. We therefore extract features from multimodal LFI representations and adopt multitask learning to jointly analyze compression distortions, enabling a holistic characterization of LFI quality degradation.

3 Proposed Database

3.1 Source Sequences

The selection of source sequences (SRCs) plays a crucial role in constructing a high-quality database. To cover diverse real-world application scenarios and improve the accuracy and generalization of subjective evaluation results, it is essential to ensure the diversity of the SRCs. In this study, the diversity of SRCs is considered from two aspects: type diversity and content diversity. In terms of type diversity, two types of SRCs are involved, including real-world scenes and synthetic scenes. Regarding content diversity, several attributes



Figure 1: Illustration of central view of SRCs.

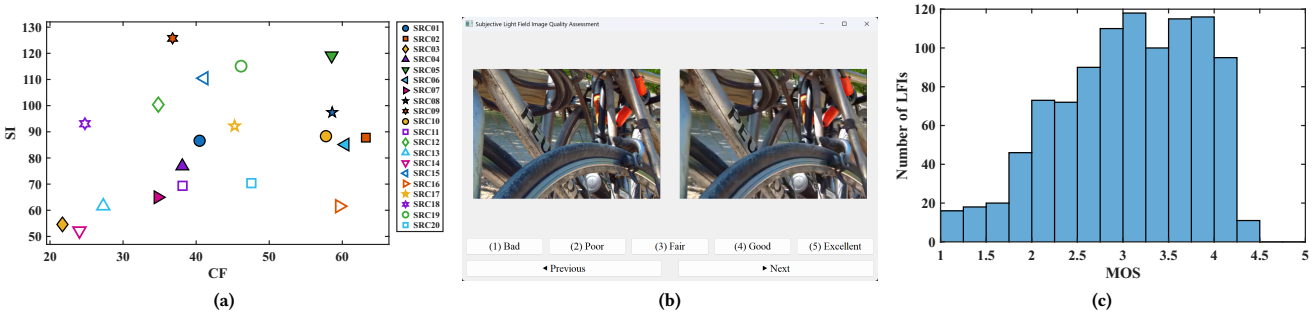
are considered when selecting SRCs, such as spatial information (SI) and colorfulness (CF).

Specifically, the SRCs of the proposed HUST-LFI database consist of 20 reference LFIs, including 10 real-world scenes [15, 32] and 10 synthetic scenes [13, 35]. Figure 1 shows the central views of all SRCs. The selected SRCs cover a wide range of visual characteristics, including different colorfulness levels (vivid and low saturation), object scales (small-scale and large-scale objects), depth ranges (near and far scenes), and scene types (indoor and outdoor), ensuring the diversity and representativeness of the dataset content. Furthermore, to objectively quantify the diversity of the source content, following [5, 57], we compute and analyze the SI and CF of the selected SRCs, whose distributions are illustrated in Figure 2a. The SI and CF values are calculated according to ITU-R BT.500 [16] and the Hasler method [11], respectively.

3.2 Hypothetical Reference Circuits

To systematically investigate the impact of different compression methods on LFI quality, the proposed database defines ten hypothetical reference circuits (HRCs). These HRCs cover multiple compression paradigms, including three traditional image and video compression methods (JPEG2000 [40], HEVC [38], and VVC [2]), four learning-based image and video compression methods (Cheng [4], ELiC [12], DCVC-FM [23], and DCVC-RT [17]), one traditional light field-specific compression method (JPEG Pleno [7]), and two learning-based light field compression methods (SADN [41] and LFIC-DRASC [9]). Each HRC is applied at five distortion levels spanning a wide perceptual quality range from nearly lossless to heavily compressed, yielding 1000 distorted LFIs in total.

Notably, existing LFIQA databases lack sufficient coverage of learning-based compression schemes, particularly distortions introduced by light field-specific methods. The proposed database addresses this gap by more faithfully reflecting LFI quality degradation in practical compression scenarios. Moreover, these compression techniques emphasize different underlying mechanisms, as detailed below.



(a) (b) (c)

Figure 2: (a) Distribution of the SI and CF values of the SRCs. (b) Graphical user interface (GUI) for subjective quality assessment. (c) Distribution of MOS values in the proposed HUST-LFI database.

Traditional image and video codecs: For JPEG2000 compression [40], LFIs are encoded based on the microlens array representation, with the compression ratio (CR) set to {25, 50, 100, 150, 200}. For video-based codecs, the light field is rearranged into a pseudo-video sequence following a serpentine scanning order, and then encoded using HEVC [38] (HM) and VVC [2] (VTM) under the low-delay-P configuration. The quantization parameter (QP) is set to {24, 29, 34, 39, 44} to control compression strength.

Learning-based image and video codecs: For learning-based compression methods, including Cheng [4] and ELiC [12], LFIs are compressed based on the microlens array structure, while DCVC-FM [23] and DCVC-RT [17] are employed to encode the pseudo-video sequences generated from LFIs. Different compression levels are achieved by selecting pretrained models with varying quality levels, corresponding to distinct rate–distortion trade-offs learned during training. Specifically, five quality levels are adopted to generate five distortion levels.

Traditional light field-specific codecs: JPEG Pleno [7], developed by the JPEG Committee, has been established as a representative standard for light field compression, providing dedicated coding architectures tailored to high-dimensional light field data. To account for distortions introduced by standardized light field compression, LFIs in the proposed database are encoded within the JPEG Pleno framework. The rate–distortion trade-off is controlled by the parameter λ, which is set to {1000, 5000, 10 000, 30 000, 50 000} to generate five distortion levels.

Learning-based light field-specific codecs: SADN [41] and LFIC-DRASC [9] are end-to-end learning-based LFI compression models. To accommodate the 9×9 angular resolution of LFIs, we retrain five models for each method using their official implementations. The rate–distortion trade-off is controlled by the parameter λ, which is set to {0.01, 0.005, 0.001, 0.0006, 0.0002}.

3.3 Subjective Experiments and Analyses

The subjective quality assessment follows the double stimulus impairment scale (DSIS) protocol recommended by ITU-R BT.500 [16]. The views of each LFI are organized into pseudo-video sequences using a serpentine scanning order. Reference and distorted sequences are randomly displayed on the left and right sides and presented simultaneously. Each sequence is played twice, after which observers rate perceptual quality using a five-point scale from 5 (excellent) to 1 (bad), as shown in Figure 2b.

To ensure experimental reliability, the setup follows the viewing conditions recommended by ITU-R BT.500 [16]. All experiments are conducted in a controlled laboratory environment using an AOC U27N3R 4K display with the resolution set to 1920×1080, allowing all LFIs to be presented at their native spatial resolution. The viewing distance is approximately seven times the image height, and ambient illumination is controlled below 20 LUX.

A total of 32 observers (18 males and 14 females, aged 21–29) participate in the study. None has prior experience in image quality assessment. Normal visual acuity (or corrected vision) and color vision are verified using the Snellen chart and Ishihara test, respectively. The experiment comprises four sessions, each containing five reference LFIs and 250 corresponding distorted LFIs. A five-minute break is provided between sessions to alleviate visual fatigue. Prior to the formal experiment, observers receive training on the scoring criteria and software operation using four additional light field contents and their distorted samples.

Outlier detection follows ITU-R BT.500 [16]. Rating statistics and their 95% confidence intervals are first computed, and observer screening is performed using the hypothesis testing method described in [6]. The ratings from three observers are discarded, and the remaining 29 qualified observers are used for subsequent statistical analysis. Subsequently, the Mean Opinion Score (MOS) of each LFI is calculated as follows.

$$MOS_j = \frac{1}{N} \sum_{i=1}^{N} m_{ij}, \tag{1}$$

where N denotes the number of observers (N=29), and m_{ij} represents the score assigned to stimulus j by observer i.

Finally, statistical analysis is conducted on the collected MOS. Figure 2c illustrates the distribution of MOS for the distorted LFIs in the proposed database. It can be observed that the quality scores are widely and reasonably distributed, further demonstrating the effectiveness of the selected distortion types and the rationality of the subjective experiment design.

4 Proposed Metric

The proposed metric consists of two key components: multi-modal patch generation, shown in Figure 3, and the MM-BLFQA quality

assessment network, illustrated in Figure 4. MM-BLFQA comprises three modules: an image-domain branch and a video-domain branch that predict preliminary LFI quality scores from the image and video representations, respectively, and a distortion-aware multi-task learning module that exploits cross-modal distortion features via two auxiliary tasks to guide the main quality assessment task, yielding the final LFI quality score.

4.1 Multi-modal Patch Generation

In many previous studies, patch-based modeling strategies have been demonstrated to be effective for LFIQA [45, 54, 55, 58]. Existing LFIQA metrics typically rely on low-dimensional representations of LFIs, such as sub-aperture images (SAIs), micro-lens images (MLIs), and pseudo video sequences (PVS), which correspond to 2D or 3D representations. However, whether such reduced representations can sufficiently capture the overall perceptual quality of 4D LFIs remains unclear. From the perspective of human visual perception, light field content inherently exhibits cross-modal characteristics shared by both images and videos, where perceived quality is jointly governed by coupled spatial and angular variations. Consequently, relying solely on single-modal 2D or 3D representations is insufficient to model the intrinsic 4D distortion characteristics of LFIs.

To this end, we introduce a multi-modal patch generation module to extract complementary multi-modal patches that more comprehensively reflect LFI perceptual quality, as illustrated in Figure 3. Input LFI is denoted as $L \in \mathbb{R}^{U\times V\times H\times W}$, where $U \times V$ and $H \times W$ represent the angular and spatial resolutions, respectively. Since the human visual system is more sensitive to luminance information [34], the original LFI, represented in the RGB color space, is first converted to the YCbCr color space. Only the luminance component Y of the central $A \times A$ SAIs is retained for patch generation, preserving perceptual information while reducing computation:

$$L_{SAI} = \{YCbCr(Central(L))\}_Y, \tag{2}$$

where $Central(\cdot)$ and $YCbCr(\cdot)$ denote the central SAI selection and YCbCr color space transformation, and A is set to 5.

To jointly capture angular and spatial information in LFIs, an $A \times A$ sliding window array is defined. Each window corresponds to a square region of size 32×32, ensuring equal spatial coverage along both horizontal and vertical directions. The sliding window array is uniformly applied over each SAI for patch extraction. At each sliding position, an $A \times A$ patch of size 32×32 is extracted, and all patches are stitched to form an SAI patch representation. Moreover, a permutation operation is applied to each SAI patch to interchange its angular and spatial coordinates, thereby generating the corresponding MLI patch L_{MLI}. Similarly, by rearranging the views within each SAI patch along the temporal dimension, a PV patch L_{PV} is constructed. An illustrative example of this permutation operation is shown in the dashed box in Figure 4. After applying the above operations to all L_{SAI}, a set of multi-modal patches are obtained, expressed as follows:

$$L_{MLI}, L_{PV} = Permute(L_{SAI}), \tag{3}$$

where $Permute(\cdot)$ is the permutation operation.

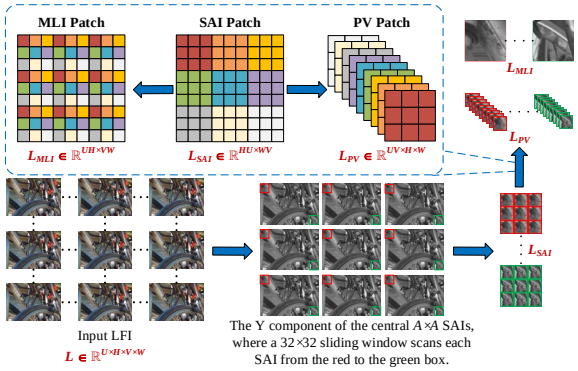


Figure 3: Preprocessing pipeline for multi-modal patch generation. For better visualization, A is set to 3 for all illustrations with $A \times A$ angular resolution.

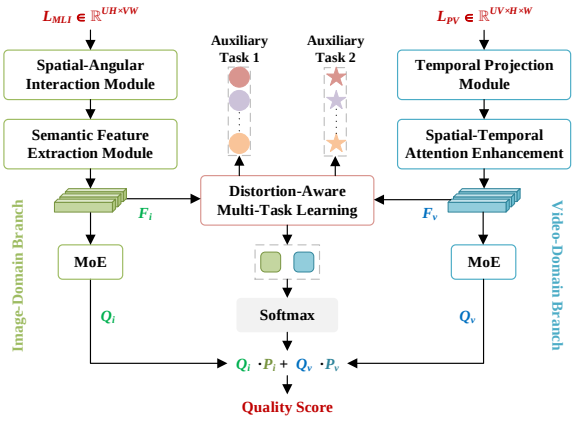


Figure 4: Overview network of the proposed MM-BLFQA.

4.2 Image-domain Branch Quality Prediction

The image-domain branch consists of two main components: a spatial-angular interaction module and a semantic feature extraction module, as illustrated in Figure 5. Given an input MLI patch $L_{MLI} \in \mathbb{R}^{1\times1\times(5\times32)\times(5\times32)}$, the spatial-angular interaction module employs spatial-angular separable convolution layers to explicitly model the intrinsic spatial-angular structure of LFIs. Specifically, the input is reshaped into a ($batchsize$, $channel$, h, w) format to enable joint interaction and fusion of spatial and angular information.

$$F_i^1 = SAIM(L_{MLI}), \tag{4}$$

where $SAIM(\cdot)$ denotes the spatial-angular interaction module, which produces intermediate features $F_i^1 \in \mathbb{R}^{1\times8\times(5\times32)\times(5\times32)}$.

The intermediate features F_i^1 are then fed into the proposed semantic feature extraction module to learn high-level spatial and angular representations. To enhance representation capacity while alleviating gradient degradation, three residual blocks are employed. Each residual block consists of two convolutional layers, two group normalization (GN) layers, and two LeakyReLU activation functions. Subsequently, a 2D average pooling layer aggregates the deep distortion-aware features into a compact representation $F_i \in \mathbb{R}^{128}$.

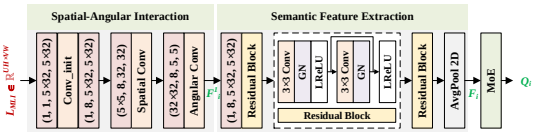


Figure 5: Architecture of the image-domain branch.

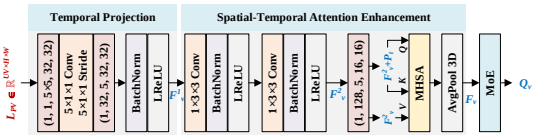


Figure 6: Architecture of the video-domain branch.

Finally, a mixture-of-experts (MoE) [49], comprising multiple collaboratively learned experts, is employed to predict the preliminary quality score Q_i from the image-domain representation of the LFI, which can be formulated as follows:

$$Q_i = MoE(SFEM(F_i^1)), \tag{5}$$

where $SFEM(\cdot)$ is the semantic feature extraction module.

4.3 Video-domain Branch Quality Prediction

The video-domain branch consists of two main components: a temporal projection module and a spatial-angular attention enhancement module, as illustrated in Figure 6. Given an input PV patch $L_{PV} \in \mathbb{R}^{1\times1\times(5\times5)\times32\times32}$, the temporal projection module employs a 3D convolution with a kernel size of 5×1×1 and a stride of 5×1×1 to project angular information into the temporal feature domain. This operation effectively removes angular redundancy while reducing the computational complexity along the temporal dimension of the feature representation. The convolution is subsequently followed by batch normalization and a LeakyReLU activation function, yielding the intermediate feature representation $F_v^1 \in \mathbb{R}^{1\times1\times5\times32\times32}$.

$$F_v^1 = TPM(L_{PV}), \tag{6}$$

where $TPM(\cdot)$ represents the temporal projection module.

The intermediate features F_v^1 are subsequently fed into the proposed spatial–temporal attention enhancement module. Within this module, a 3D ConvBlock with a dedicated structural design is first employed to extract deep representations, producing the intermediate features F_v^2. A temporal positional encoding P_t, defined along the temporal dimension, is then added to F_v^2, followed by layer normalization (LN) to generate the query Q, key K, and value V representations:

$$Q = K = LN(F_v^2 + P_t), \quad V = F_v^2. \tag{7}$$

Thereafter, multi-head self-attention (MHSA) is applied to capture the dependencies among temporal tokens. The output tokens are subsequently passed through a feed-forward network (FFN), resulting in enhanced feature representations. The overall process can be summarized as follows:

$$F_v^{2,att} = MHSA(Q, K, V) + F_v^2, \tag{8}$$

$$\hat{F}_v^{2,att} = FFN(F_v^{2,att}) + F_v^{2,att}. \tag{9}$$

Finally, a 3D average pooling layer aggregates the deep distortion-aware features into a compact representation $F_v \in \mathbb{R}^{128}$, which is fed into a MoE module to predict the preliminary video-domain quality score Q_v:

$$Q_v = MoE(F_v). \tag{10}$$

4.4 Distortion-aware Multi-task Learning

The distortion-aware multi-task learning module consists of two components: auxiliary distortion prediction tasks and the main quality assessment task.

(1) Auxiliary distortion prediction. Multi-task learning has been widely demonstrated to enhance the performance of the main task through the joint optimization of related auxiliary tasks [27, 59]. Considering that distortion type and distortion level play critical roles in subjective quality perception, two auxiliary tasks are introduced, including distortion type discrimination and distortion level classification, to enhance the representation of distortion attributes. Specifically, each auxiliary task is implemented using an independent fully connected (FC) layer, whose output dimension equals the number of classes of the corresponding task. In the constructed database, the numbers of distortion types and distortion levels are 10 and 5, respectively.

(2) Main quality assessment. Single-modal methods based solely on image-domain or video-domain representations suffer from inherent limitations, as they are unable to effectively capture cross-modal interactions or fully model multi-modal distortion characteristics [25], thereby restricting their performance in 4D LFI quality assessment. To address this issue, cross-modal relationships are modeled based on shared distortion-domain similarity. Specifically, a distortion-domain classifier is first employed to model the fused multi-modal features and output the probability distribution indicating whether the input belongs to the image distortion domain or the video distortion domain. These probabilities are then used as adaptive weights to fuse the quality predictions (Q_i and Q_v) from different modality branches, yielding the final predicted LFI quality score Q, which is formulated as follows:

$$F_{i,v} = Concat(F_i, F_v), \tag{11}$$

$$[P_i, P_v] = Softmax(FC(F_{i,v})), \tag{12}$$

$$Q = Q_i \cdot P_i + Q_v \cdot P_v, \tag{13}$$

where $F_{i,v}$ denotes the cross-modal distortion feature, and P_i and P_v are adaptive weights determined by distortion-domain similarity.

4.5 Loss Function

The mean squared error loss and cross-entropy loss are employed to optimize the main quality assessment task and the auxiliary distortion prediction tasks, respectively, which are defined as follows:

$$L_q = \frac{1}{B}\sum_{b=1}^{B}(\hat{p}_b - p_b)^2, \tag{14}$$

$$L_d = -\frac{1}{B}\sum_{b=1}^{B}\sum_{d=1}^{D} p_{b,d}\log(\hat{p}_{b,d}), \tag{15}$$

$$L_c = -\frac{1}{B}\sum_{b=1}^{B}\sum_{c=1}^{C} p_{b,c}\log(\hat{p}_{b,c}), \tag{16}$$

where B denotes the batch size. $\hat{p}_b$ and p_b represent the predicted quality score and the corresponding MOS. $\hat{p}_{b,d}$ and $p_{b,d}$ denote the predicted and ground-truth distortion types, while $\hat{p}_{b,c}$ and $p_{b,c}$ denote the predicted and ground-truth distortion levels.

To jointly optimize the parameters of the three learning tasks, the overall loss function is formulated as:

$$L = L_q + \lambda_d \cdot L_d + \lambda_c \cdot L_c, \tag{17}$$

where λ_d and λ_c denote the weighting coefficients that balance the contributions of the auxiliary tasks. In this work, they are empirically set to 0.1 and 0.1, respectively.

5 Experiments

5.1 Experimental Setups

5.1.1 Implementation Details. Experiments are conducted on a system with an Intel i9-13900KF CPU, NVIDIA GeForce RTX 3090 GPU, and 32GB RAM under Ubuntu 20.04. Training uses mini-batch SGD with momentum 0.9, weight decay 0.001, an initial learning rate of 0.001, and batch size 32, decayed by a factor of 0.1 every 30 epochs, for a total of 70 epochs.

5.1.2 Train-test Splitting. A standard five-fold cross-validation protocol is adopted on the proposed database. Specifically, the 20 scenes are randomly divided into 5 mutually exclusive subsets, each containing 4 scenes, ensuring complete independence between training and test sets. In each fold, the 200 distorted LFIs from the corresponding subset serve as the test set, while the remaining 800 are used for training. The reported results represent the average performance across all folds.

5.1.3 Evaluation Criteria. Four evaluation criteria are used to quantify the correlation between predicted scores and MOS values: pearson linear correlation coefficient (PLCC), spearman rank-order correlation coefficient (SRCC), kendall rank correlation coefficient (KRCC), and root mean square error (RMSE). Higher values of PLCC, SRCC, KRCC, along with lower RMSE values, indicate better performance. Prior to PLCC and RMSE computation, predicted scores are rescaled to the MOS range via a nonlinear logistic function [10].

5.1.4 Compared Metrics. To validate the effectiveness of the proposed metric, we compare it with 15 blind IQA metrics, including two handcrafted 2DIQA metrics (BRISQUE [29] and GLBP [24]), three deep learning-based 2DIQA metrics (MANIQA [50], VCRNet [30], and LoDa [48]), six handcrafted LFIQA metrics (NR-LFQA [37], Tensor-NLFQ [60], PVRI [47], 4D-DCT-LFIQA [46], SAB [53], and MPT-MWI-BLiF [52]), and four deep learning-based LFIQA metrics (DeeBLiF [54], PVBLiF [55], FSFNet [45], and MAFBLiF [58]). For a fair comparison, all compared metrics are retrained and evaluated on the proposed database using the official implementations provided by authors under the same hardware configuration.

5.2 Performance Comparison

5.2.1 Overall Performance Comparison. We benchmark the proposed metric against representative blind 2DIQA and blind LFIQA metrics on the proposed database. As summarized in Table 2, where the best and second-best results are highlighted in bold and underlined, our metric consistently achieves superior overall performance across all evaluation criteria.

Table 2: Performance comparison on the proposed database.

Type	Metric	PLCC↑	SRCC↑	KRCC↑	RMSE↓
2DIQA	BRISQUE [29]	0.5336	0.4960	0.3583	0.6184
	GLBP [24]	0.5445	0.4555	0.3217	0.6195
	MANIQA [50]	0.6004	0.5223	0.3858	0.5796
	VCRNet [30]	0.6272	0.5595	0.4083	0.5685
	LoDa [48]	0.6096	0.5512	0.3997	0.5935
LFIQA	NR-LFQA [37]	0.7179	0.6867	0.5028	0.5229
	Tensor-NLFQ [60]	0.6605	0.6373	0.4669	0.5544
	PVRI [47]	0.6760	0.6293	0.4661	0.5344
	4D-DCT-LFIQA [46]	0.6749	0.6333	0.4626	0.5508
	SAB [53]	0.7129	0.6408	0.4678	0.5283
	MPT-MWI-BLiF [52]	0.7830	0.7488	0.5723	0.4507
	DeeBLiF [54]	0.8231	0.7948	0.6116	0.4202
	PVBLiF [55]	0.8363	<u>0.8077</u>	<u>0.6295</u>	<u>0.4016</u>
	FSFNet [45]	0.8283	0.7935	0.6075	0.4167
	MAFBLiF [58]	<u>0.8417</u>	0.8047	0.6156	0.4030
	Proposed	**0.8724**	**0.8572**	**0.6821**	**0.3586**

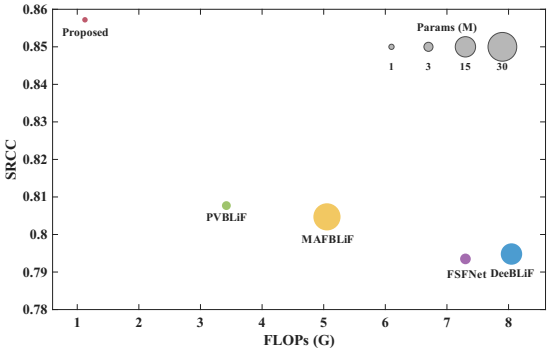


Figure 7: Comparison of computational complexity and quality prediction performance among competing LFIQA metrics.

5.2.2 Comparison of Different Distortion Types. We further evaluate the robustness of all metrics across different distortion types, with SRCC results reported in Table 3. The hit-count denotes the number of top-two rankings each metric achieves across all distortion types. As shown, the proposed metric attains nine top-two rankings out of ten distortion types, demonstrating strong robustness to diverse compression artifacts.

5.2.3 Computational Complexity. As shown in Figure 7, we compare the proposed metric with competing deep learning-based LFIQA metrics in terms of FLOPs and model parameter size. The proposed metric achieves top prediction performance while maintaining low computational complexity, striking an excellent efficiency-accuracy tradeoff.

5.2.4 Statistical Significance Test. To further validate statistical significance, we conduct a t-test on SRCC and PLCC values of the proposed and competing LFIQA metrics on the proposed database. As shown in Figure 8, "-1", "0", and "1" indicate statistically worse, competitive, and superior performance, respectively. The proposed metric consistently outperforms all LFIQA metrics.

Table 3: Comparison of SRCC performance of different distortion types on the proposed database.

Type	Metric	JPEG 2000	HEVC	VVC	Cheng	ELiC	DCVC-FM	DCVC-RT	JPEG Pleno	SADN	LFIC-DRASC	Hit-count
2DIQA	BRISQUE [29]	0.5722	0.6264	0.5759	0.5390	0.6720	0.4529	0.4577	0.4323	0.3728	0.4141	0
	GLBP [24]	0.4878	0.6037	0.6280	0.3267	0.4355	0.4474	0.4972	0.5030	0.4502	0.3880	0
	MANIQA [50]	0.5725	0.7949	0.8066	0.5187	0.6055	0.7903	0.7618	0.5706	0.3818	0.3882	0
	VCRNet [30]	0.5393	0.7778	0.7770	0.5154	0.5865	0.6107	0.5920	0.5776	0.3802	0.4336	0
	LoDa [48]	0.4418	0.6954	0.7314	0.4326	0.6244	0.5967	0.5582	0.6811	0.4183	0.3899	0
LFIQA	NR-LFQA [37]	0.6891	0.7879	0.7907	0.6128	0.7247	0.7822	0.7089	0.8126	0.5143	0.5162	0
	Tensor-NLFQ [60]	0.6300	0.7382	0.7111	0.5732	0.5900	0.6674	0.6459	0.6758	0.7500	**0.6951**	1
	PVRI [47]	0.4541	0.7870	0.7790	0.5623	0.6410	0.6740	0.6868	0.7172	0.4552	0.3893	0
	4D-DCT-LFIQA [46]	0.4836	0.7995	0.7836	0.4967	0.5418	0.7749	0.7234	0.6178	0.4619	0.5212	0
	SAB [53]	0.7184	0.7940	0.8225	0.4819	0.7616	0.6618	0.6927	0.6461	0.4139	0.4857	0
	MPT-MWI-BLiF [52]	0.6084	0.8760	0.8737	0.6309	0.7697	0.8102	0.8034	0.7081	0.6249	0.5982	0
	DeeBLiF [54]	0.7817	0.9147	0.9043	0.6836	0.7928	0.8749	0.8562	0.7571	**0.6978**	0.6375	2
	PVBLiF [55]	0.8235	0.9304	0.9038	0.7223	0.8085	0.8185	0.8200	0.7852	0.6970	0.6310	2
	FSFNet [45]	0.8223	0.8983	0.8877	0.7254	0.8355	0.8417	0.8219	0.8254	0.6883	0.6032	0
	MAFBLiF [58]	0.8189	0.9319	0.9269	0.7650	0.8496	0.8542	0.8581	0.8618	0.6425	0.5914	6
	Proposed	**0.8605**	**0.9364**	**0.9314**	**0.7968**	**0.8832**	**0.9103**	**0.8855**	**0.8832**	0.6862	0.6694	**9**

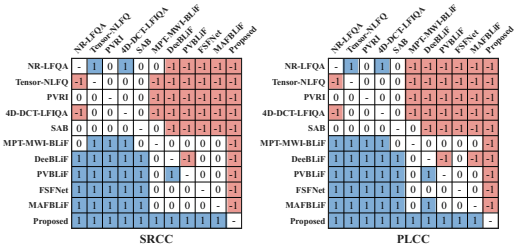


Figure 8: Statistical significance between LFIQA metrics.

Table 4: Cross-database validation for the proposed metric.

Database	PLCC↑	SRCC↑	KRCC↑	RMSE↓
Win5-LID [36]	0.7712	0.7202	0.5618	0.6081
SHU [33]	0.7831	0.7464	0.5616	0.6752

5.3 Cross-database Validation

To validate the generalization capability of the proposed metric, cross-database experiments are conducted by training on the proposed database and testing on Win5-LID [36] and SHU [33] databases. As shown in Table 4, the performance does not drop significantly, indicating the strong generalization ability of the proposed metric.

5.4 Ablation Studies

To evaluate the contribution of each module in MM-BLFQA, we conduct ablation studies summarized in Table 5. Removing the spatial-angular interaction module (SAIM) causes a significant performance drop, indicating that spatial and angular structures are interdependent. Similarly, removing the distortion-aware multi-task learning module (DMLM) degrades performance, confirming the effectiveness of cross-modal distortion-aware features. The full model achieves the best predictive performance.

We further investigate the contribution of each branch through single-modal experiments. As shown in Table 6, the 2D image and 3D video representations capture complementary dimensional information inherent in 4D LFIs, and their fusion yields the most comprehensive distortion representation and best performance.

Table 5: Ablation study of MM-BLFQA modules.

Module	PLCC↑	SRCC↑	KRCC↑	RMSE↓
w/o SAIM	0.8375	0.8141	0.6328	0.3984
w/o SFEM	0.8598	0.8426	0.6664	0.3696
w/o TPM	0.8638	0.8412	0.6567	0.3775
w/o STAEM	0.8589	0.8368	0.6514	0.3790
w/o DMLM	0.8431	0.8208	0.6367	0.3973
All	**0.8724**	**0.8572**	**0.6821**	**0.3586**

Table 6: Ablation study on single-modal branches.

Branch	PLCC↑	SRCC↑	KRCC↑	RMSE↓
image	0.8241	0.7896	0.6006	0.4241
video	0.8213	0.7929	0.6134	0.4131
All	**0.8724**	**0.8572**	**0.6821**	**0.3586**

6 Conclusion

This work investigates perceptual quality assessment for light field image compression. To this end, we construct a subjective database covering distortions introduced by traditional and learning-based image, video, and light field-specific compression methods. The database offers advantages in scale, content diversity, and distortion types, providing a systematic and comprehensive foundation for analyzing perceptual artifacts in compressed light field content. Based on this, we propose a blind quality assessment metric that leverages multi-modal and multi-task learning to efficiently predict quality scores. Experimental results demonstrate that the proposed metric outperforms existing representative quality assessment metrics in various aspects while maintaining low computational complexity.

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References

[1] Waqas Ahmad, Roger Olsson, and Mårten Sjöström. 2017. Interpreting plenoptic images as multi-view sequences for improved compression. In *2017 IEEE International Conference on Image Processing (ICIP)*. 4557–4561.

[2] Benjamin Bross, Ye-Kui Wang, Yan Ye, Shan Liu, Jianle Chen, Gary J. Sullivan, and Jens-Rainer Ohm. 2021. Overview of the Versatile Video Coding (VVC) Standard and its Applications. *IEEE Transactions on Circuits and Systems for Video Technology* 31, 10 (2021), 3736–3764.

[3] Yu Chen, Binbin Yan, Shuo Chen, and Xinzhu Sang. 2025. A Comprehensive Model for Visual Fatigue Assessment in 3D Light Field Displays Based on Eye Movement Data Analysis. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*. 7036–7044.

[4] Zhengxue Cheng, Heming Sun, Masaru Takeuchi, and Jiro Katto. 2020. Learned Image Compression With Discretized Gaussian Mixture Likelihoods and Attention Modules. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 7936–7945.

[5] Yueli Cui, Gangyi Jiang, Mei Yu, Yeyao Chen, and Yo-Sung Ho. 2024. Stitched Wide Field of View Light Field Image Quality Assessment: Benchmark Database and Objective Metric. *IEEE Transactions on Multimedia* 26 (2024), 5092–5107.

[6] Francesca De Simone, Lutz Goldmann, Vittorio Baroncini, and Touradj Ebrahimi. 2009. Subjective evaluation of JPEG XR image compression. In *Applications of Digital Image Processing XXXII*, Vol. 7443. SPIE, 177–188.

[7] Touradj Ebrahimi, Siegfried Foessel, Fernando Pereira, and Peter Schelkens. 2016. JPEG Pleno: Toward an Efficient Representation of Visual Reality. *IEEE MultiMedia* 23, 4 (2016), 14–20.

[8] Yihui Fan, Xin Jin, Siyao Zhou, and Shun Zou. 2024. Light Fields Stitching for Windowed-6DoF VR Content. *IEEE Transactions on Circuits and Systems for Video Technology* 34, 9 (2024), 8736–8752.

[9] Shiyu Feng, Yun Zhang, Linwei Zhu, and Sam Kwong. 2025. LFIC-DRASC: Deep Light Field Image Compression Using Disentangled Representation and Asymmetrical Strip Convolution. *IEEE Transactions on Broadcasting* 71, 3 (2025), 889–902.

[10] Video Quality Experts Group et al. 2000. Final report from the video quality experts group on the validation of objective models of video quality assessment. In *VQEG meeting, Ottawa, Canada, March, 2000*.

[11] David Hasler and Sabine E Suesstrunk. 2003. Measuring colorfulness in natural images. In *Human vision and electronic imaging VIII*, Vol. 5007. SPIE, 87–95.

[12] Dailan He, Ziming Yang, Weikun Peng, Rui Ma, Hongwei Qin, and Yan Wang. 2022. ELIC: Efficient Learned Image Compression with Unevenly Grouped Space-Channel Contextual Adaptive Coding. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5708–5717.

[13] Katrin Honauer, Ole Johannsen, Daniel Kondermann, and Bastian Goldluecke. 2017. A Dataset and Evaluation Methodology for Depth Estimation on 4D Light Fields. In *Computer Vision – ACCV 2016*. 19–34.

[14] Zhijiao Huang, Mei Yu, Gangyi Jiang, Ken Chen, Zongju Peng, and Fen Chen. 2019. Reconstruction Distortion Oriented Light Field Image Dataset for Visual Communication. In *2019 International Symposium on Networks, Computers and Communications (ISNCC)*. 1–5.

[15] INRIA. 2026. INRIA Lytro Image Dataset. https://www.irisa.fr/temics/demos/lightField/LowRank2/datasets/datasets.html.

[16] B. T. ITU-R. 2002. Methodology for the subjective assessment of the quality of television pictures. https://www.itu.int/rec/R-REC-BT.500.

[17] Zhaoyang Jia, Bin Li, Jiahao Li, Wenxuan Xie, Linfeng Qi, Houqiang Li, and Yan Lu. 2025. Towards Practical Real-Time Neural Video Compression. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 12543–12552.

[18] Jing Jin, Junhui Hou, Hui Yuan, and Sam Kwong. 2020. Learning light field angular super-resolution via a geometry-aware network. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 34. 11141–11148.

[19] Nima Khademi Kalantari, Ting-Chun Wang, and Ravi Ramamoorthi. 2016. Learning-based view synthesis for light field cameras. *ACM Trans. Graph.* 35, 6 (2016), 10 pages.

[20] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkuehler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Trans. Graph.* 42, 4 (2023).

[21] Jiwon Kim, Jung Kwon Lee, and Kyoung Mu Lee. 2016. Accurate Image Super-Resolution Using Very Deep Convolutional Networks. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 1646–1654.

[22] Marc Levoy and Pat Hanrahan. 1996. Light field rendering. In *Proceedings of the 23rd Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '96)*. 31–42.

[23] Jiahao Li, Bin Li, and Yan Lu. 2024. Neural Video Compression with Feature Modulation. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 26099–26108.

[24] Qiaohong Li, Weisi Lin, and Yuming Fang. 2016. No-Reference Quality Assessment for Multiply-Distorted Images in Gradient Domain. *IEEE Signal Processing Letters* 23, 4 (2016), 541–545.

[25] Junhao Lin, Yueli Cui, Chenli Fang, Binghong Pan, Chencheng Pan, Gangyi Jiang, Shiqing Zhang, Siwei Ma, and Qi Tian. 2026. Multi-modal Cross-Attention Guided Network for Audio-Visual Quality Evaluation via Visual Saliency and Mel-spectrum Features. *IEEE Transactions on Circuits and Systems for Video Technology* (2026), 1–1.

[26] Gaosheng Liu, Huanjing Yue, Jiamin Wu, and Jingyu Yang. 2021. Intra-inter view interaction network for light field image super-resolution. *IEEE Transactions on Multimedia* 25 (2021), 256–266.

[27] Yun Liu, Sifan Li, Huiyu Duan, Yu Zhou, Daoxin Fan, and Guangtao Zhai. 2025. Multi-Task Guided No-Reference Omnidirectional Image Quality Assessment With Feature Interaction. *IEEE Transactions on Circuits and Systems for Video Technology* 35, 9 (2025), 8794–8806.

[28] Nan Meng, Hayden K.-H. So, Xing Sun, and Edmund Y. Lam. 2021. High-Dimensional Dense Residual Convolutional Neural Network for Light Field Reconstruction. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 43, 3 (2021), 873–886.

[29] Anish Mittal, Anush Krishna Moorthy, and Alan Conrad Bovik. 2012. No-Reference Image Quality Assessment in the Spatial Domain. *IEEE Transactions on Image Processing* 21, 12 (2012), 4695–4708.

[30] Zhaoqing Pan, Feng Yuan, Jianjun Lei, Yuming Fang, Xiao Shao, and Sam Kwong. 2022. VCRNet: Visual Compensation Restoration Network for No-Reference Image Quality Assessment. *IEEE Transactions on Image Processing* 31 (2022), 1613–1627.

[31] Pradip Paudyal, Federica Battisti, Mårten Sjöström, Roger Olsson, and Marco Carli. 2017. Towards the Perceptual Quality Evaluation of Compressed Light Field Images. *IEEE Transactions on Broadcasting* 63, 3 (2017), 507–522.

[32] Martin Rerabek and Touradj Ebrahimi. 2016. New light field image dataset. In *8th international conference on quality of multimedia experience (QoMEX)*.

[33] Liang Shan, Ping An, Chunli Meng, Xinpeng Huang, Chao Yang, and Liquan Shen. 2019. A No-Reference Image Quality Assessment Metric by Multiple Characteristics of Light Field Images. *IEEE Access* 7 (2019), 127217–127229.

[34] Xiwu Shang, Jie Liang, Guozhong Wang, Haiwu Zhao, Chengjia Wu, and Chang Lin. 2019. Color-Sensitivity-Based Combined PSNR for Objective Video Quality Assessment. *IEEE Transactions on Circuits and Systems for Video Technology* 29, 5 (2019), 1239–1250.

[35] Jinglei Shi, Xiaoran Jiang, and Christine Guillemot. 2019. A Framework for Learning Depth From a Flexible Subset of Dense and Sparse Light Field Views. *IEEE Transactions on Image Processing* 28, 12 (2019), 5867–5880.

[36] Likun Shi, Shengyang Zhao, Wei Zhou, and Zhibo Chen. 2018. Perceptual Evaluation of Light Field Image. In *2018 25th IEEE International Conference on Image Processing (ICIP)*. 41–45.

[37] Likun Shi, Wei Zhou, Zhibo Chen, and Jinglin Zhang. 2020. No-Reference Light Field Image Quality Assessment Based on Spatial-Angular Measurement. *IEEE Transactions on Circuits and Systems for Video Technology* 30, 11 (2020), 4114–4128.

[38] Gary J. Sullivan, Jens-Rainer Ohm, Woo-Jin Han, and Thomas Wiegand. 2012. Overview of the High Efficiency Video Coding (HEVC) Standard. *IEEE Transactions on Circuits and Systems for Video Technology* 22, 12 (2012), 1649–1668.

[39] Ioan Tabus, Petri Helin, and Pekka Astola. 2017. Lossy compression of lenslet images from plenoptic cameras combining sparse predictive coding and JPEG 2000. In *2017 IEEE International Conference on Image Processing (ICIP)*. 4567–4571.

[40] D.S. Taubman and M.W. Marcellin. 2002. JPEG2000: standard for interactive imaging. *Proc. IEEE* 90, 8 (2002), 1336–1357.

[41] Kedeng Tong, Xin Jin, Chen Wang, and Fan Jiang. 2022. SADN: Learned Light Field Image Compression with Spatial-Angular Decorrelation. In *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 1870–1874.

[42] Irene Viola and Touradj Ebrahimi. 2018. VALID: Visual quality Assessment for Light field Images Dataset. In *2018 Tenth International Conference on Quality of Multimedia Experience (QoMEX)*. 1–3.

[43] Yingqian Wang, Longguang Wang, Gaochang Wu, Jungang Yang, Wei An, Jingyi Yu, and Yulan Guo. 2023. Disentangling Light Fields for Super-Resolution and Disparity Estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 1 (2023), 425–443.

[44] Gaochang Wu, Mandan Zhao, Liangyong Wang, Qionghai Dai, Tianyou Chai, and Yebin Liu. 2017. Light Field Reconstruction Using Deep Convolutional Network

on EPI. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 1638–1646.

[45] Jianjun Xiang, Yuanjie Dang, Peng Chen, Ronghua Liang, Ruohong Huan, and Zhengyu Zhang. 2023. Spatial-angular Quality-aware Representation Learning for Blind Light Field Image Quality Assessment. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*. 1077–1087.

[46] Jianjun Xiang, Gangyi Jiang, Mei Yu, Zhidi Jiang, and Yo-Sung Ho. 2023. No-Reference Light Field Image Quality Assessment Using Four-Dimensional Sparse Transform. *IEEE Transactions on Multimedia* 25 (2023).

[47] Jianjun Xiang, Mei Yu, Gangyi Jiang, Haiyong Xu, Yang Song, and Yo-Sung Ho. 2021. Pseudo Video and Refocused Images-Based Blind Light Field Image Quality Assessment. *IEEE Transactions on Circuits and Systems for Video Technology* 31, 7 (2021), 2575–2590.

[48] Kangmin Xu, Liang Liao, Jing Xiao, Chaofeng Chen, Haoning Wu, Qiong Yan, and Weisi Lin. 2024. Boosting Image Quality Assessment Through Efficient Transformer Adaptation with Local Feature Enhancement. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2662–2672.

[49] Junfeng Yang, Jing Fu, Wei Zhang, Wenzhi Cao, Limei Liu, and Han Peng. 2024. MoE-AGIQA: Mixture-of-Experts Boosted Visual Perception-Driven and Semantic-Aware Quality Assessment for AI-Generated Images. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. 6395–6404.

[50] Sidi Yang, Tianhe Wu, Shuwei Shi, Shanshan Lao, Yuan Gong, Mingdeng Cao, Jiahao Wang, and Yujiu Yang. 2022. MANIQA: Multi-Dimension Attention Network for No-Reference Image Quality Assessment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*. 1191–1200.

[51] Shuo Zhang, Hao Sheng, Da Yang, Jun Zhang, and Zhang Xiong. 2018. Micro-Lens-Based Matching for Scene Recovery in Lenslet Cameras. *IEEE Transactions on Image Processing* 27, 3 (2018), 1060–1075.

[52] Zhengyu Zhang, Shishun Tian, Jianjun Xiang, Wenbin Zou, Luce Morin, and Lu Zhang. 2025. Blind Light Field Image Quality Assessment Using Multiplane Texture and Multilevel Wavelet Information. *IEEE Transactions on Broadcasting* 71, 4 (2025), 1092–1107.

[53] Zhengyu Zhang, Shishun Tian, Jinjia Zhou, Luce Morin, and Lu Zhang. 2025. A New Benchmark Database and Objective Metric for Light Field Image Quality Evaluation. *IEEE Transactions on Circuits and Systems for Video Technology* 35, 3 (2025), 2382–2397.

[54] Zhengyu Zhang, Shishun Tian, Wenbin Zou, Luce Morin, and Lu Zhang. 2022. Deeblif: Deep Blind Light Field Image Quality Assessment by Extracting Angular and Spatial Information. In *2022 IEEE International Conference on Image Processing (ICIP)*. 2266–2270.

[55] Zhengyu Zhang, Shishun Tian, Wenbin Zou, Luce Morin, and Lu Zhang. 2023. PVBLiF: A Pseudo Video-Based Blind Quality Assessment Metric for Light Field Image. *IEEE Journal of Selected Topics in Signal Processing* 17, 6 (2023), 1193–1207.

[56] Shengyang Zhao and Zhibo Chen. 2017. Light field image coding via linear approximation prior. In *2017 IEEE International Conference on Image Processing (ICIP)*. 4562–4566.

[57] Rui Zhou, Gangyi Jiang, Yeyao Chen, Yueli Cui, Ting Luo, and Haiyong Xu. 2026. Benchmark Database and Objective Metric for Wide-Field-of-View Light Field Image Quality Measurement. *IEEE Transactions on Instrumentation and Measurement* 75 (2026), 1–15.

[58] Rui Zhou, Gangyi Jiang, Yueli Cui, Yeyao Chen, Haiyong Xu, Ting Luo, and Mei Yu. 2024. MAFBLiF: Multi-Scale Attention Feature Fusion-Based Blind Light Field Image Quality Assessment. *IEEE Transactions on Broadcasting* 70, 4 (2024), 1266–1278.

[59] Rui Zhou, Gangyi Jiang, Linwei Zhu, Yeyao Chen, Yueli Cui, Ting Luo, and Haiyong Xu. 2025. Mamba-Based Blind Stitched Wide Field of View Light Field Image Quality Assessment via Dual-Viewport Sampling. In *2025 IEEE International Conference on Multimedia and Expo (ICME)*. 1–6.

[60] Wei Zhou, Likun Shi, Zhibo Chen, and Jinglin Zhang. 2020. Tensor Oriented No-Reference Light Field Image Quality Assessment. *IEEE Transactions on Image Processing* 29 (2020), 4070–4084.